

Exercise (2.3.5). Find the number of generators for $\mathbb{Z}/49000\mathbb{Z}$.

Solution: As we recalled in the previous problem, every residual class whose representative is coprime to the order of the group generates the group. Using Euler's φ – *function* we can find the number of elements from $[49000]$ which are coprime to 49000. Determining the prime factorization of 49000,

$$49000 = 49(10^3) = (7^2)(5^3)(2^3)$$

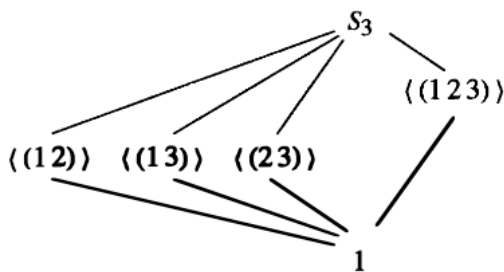
Applying Euler's φ – *function* we get,

$$\varphi(49000) = 7(6)5^2(4)2^2(1) = 16800.$$

□

Exercise (2.5.6c). Use the given lattices to help find the centralizer of every element in S_3

FIGURE 1. Lattice Diagram for S_3



Solution: Clearly $C_{S_3}(1) = S_3$. Seeing how the centralizer of an element $x \in S_3$ is a subgroup of S_3 containing x , by the way the lattice diagram is structured, $C_{S_3}(\langle x \rangle)$ must be equal to either $\langle x \rangle$ or S_3 .

Consider (12) and note that $(13)(12)(13) = (23)$, therefore $(13) \notin C_{S_3}(\langle (12) \rangle)$. Thus $C_{S_3}(\langle (12) \rangle) = \langle (12) \rangle$.

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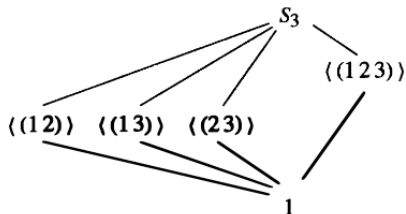
Consider (23) and note that $(13)(23)(13) = (12)$, therefore $(13) \notin C_{S_3}(\langle(23)\rangle)$. Thus $C_{S_3}(\langle(23)\rangle) = \langle(23)\rangle$.

Consider (132) and note that $(13)(132)(13) = (123)$, therefore $(13) \notin C_{S_3}(\langle(132)\rangle)$. Since $(132) = (123)^2$, $C_{S_3}(\langle(132)\rangle) = C_{S_3}(\langle(123)\rangle) = \langle(123)\rangle$. $\square$

Exercise (2.5.8 - Stefano Fochesatto).

- (1) Find the normalizer of each subgroup in S_3 .

FIGURE 2. Lattice Diagram for S_3



Solution: Clearly $N_{S_3}\langle 1 \rangle = S_3$ and $N_{S_3}\langle S_3 \rangle = S_3$. Similarly to the previous problem, since the normalizer of subgroup contains the subgroup, and because of the structure of the lattice diagram for any subgroup G , either $N_{S_3}(G) = G$ or $N_{S_3}(G) = S_3$.

Consider $\langle 12 \rangle$ and note that $(13)(12)(13) = (23)$. Since $(23) \notin \langle(12)\rangle$ we know that $(13) \notin N_{S_3}(\langle(12)\rangle)$ and therefore $N_{S_3}(\langle(12)\rangle) = \langle(12)\rangle$.

Consider $\langle 13 \rangle$ and note that $(12)(13)(12) = (23)$. Since $(23) \notin \langle(13)\rangle$ we know that $(12) \notin N_{S_3}(\langle(13)\rangle)$ and therefore $N_{S_3}(\langle(13)\rangle) = \langle(13)\rangle$.

Consider $\langle 23 \rangle$ and note that $(13)(23)(13) = (12)$. Since $(12) \notin \langle(23)\rangle$ we

know that $(13) \notin N_{S_3}(\langle(23)\rangle)$ and therefore $N_{S_3}(\langle(23)\rangle) = \langle(23)\rangle$.

Finally consider $\langle(123)\rangle = \{1, (123), (132)\}$, and note that

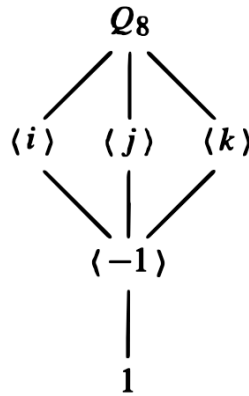
$$(12)(123)(12) = (132),$$

$$(12)(132)(12) = (123).$$

Therefore, $(12) \in N_{S_3}(\langle(123)\rangle)$, and since $(12) \notin \langle(123)\rangle$ we know that $N_{S_3}(\langle(123)\rangle) = S_3$. □

(2) Find the normalizer of each subgroup in Q_8

FIGURE 3. Lattice Diagram for Q_8



Solution: Clearly $N_{Q_8}(\langle 1 \rangle) = Q_8$ and $N_{Q_8}(\langle Q_8 \rangle) = Q_8$. Let $a \in Q_8$ and note that, $a(-1)a^{-1} = a(-1)(-1)a = a^2 = -1 \in \langle -1 \rangle$. Thus $N_{Q_8}(\langle -1 \rangle) = Q_8$. Since the normalizer of subgroup contains the subgroup, and because of the structure of the lattice diagram for subgroups $G = \{\langle i \rangle, \langle j \rangle, \langle k \rangle\}$, either $N_{Q_8}(G) = G$ or $N_{Q_8}(G) = Q_8$.

Consider $\langle i \rangle$ and note that $ji(-j) = (-k)(-j) = -i \in \langle i \rangle$ and $j - i(-j) = (j)(k) = i \in \langle i \rangle$. Therefore $j \in N_{Q_8}(\langle i \rangle)$, thus $N_{Q_8}(\langle i \rangle) = Q_8$.

Consider $\langle j \rangle$ and note that $ij(-i) = (k)(-i) = -j \in \langle j \rangle$ and $i - j(-i) = (i)(-k) = j \in \langle j \rangle$. Therefore $j \in N_{Q_8}(\langle j \rangle)$, thus $N_{Q_8}(\langle j \rangle) = Q_8$.

Consider $\langle k \rangle$ and note that $ik(-i) = (-j)(-i) = -k \in \langle k \rangle$ and $i - k(-i) = (i)(j) = k \in \langle k \rangle$. Therefore $j \in N_{Q_8}(\langle k \rangle)$, thus $N_{Q_8}(\langle k \rangle) = Q_8$. $\square$

Exercise (3.1.24 - Stefano Fochesatto). Prove that if N is a normal subgroup of G and H is any subgroup of G then $N \cap H$ is a normal subgroup of H .

Proof: Suppose N is a normal subgroup of G and H is any subgroup of G . Let $a, b \in N \cap H$, clearly $ab^{-1} \in N \cap H$, hence $N \cap H \leq H$. Note that since N is normal in G and H is a subgroup for all $h \in H$ $hah^{-1} \in N$. Since $a \in H$ and H is a subgroup we know that $hah^{-1} \in H$. Thus $hah^{-1} \in N \cap H$ and therefore $N \cap H$ is normal in H . $\square$

Exercise (3.1.36 - Stefano Fochesatto). Prove that if $G/Z(G)$ is cyclic then G is abelian. [If $G/Z(G)$ is cyclic with generator $xZ(G)$, show that every element of G can be written in the form $x^a z$ for some integer $a \in \mathbb{Z}$ and some element $z \in Z(G)$.]

Proof: Suppose $G/Z(G) = \langle xZ(G) \rangle$ for some fixed $x \in G$. Since $G/Z(G)$ is cyclic there exists some $a \in \mathbb{Z}$ such that for all $g \in G$,

$$gZ(G) = (xZ(G))^a = x^a Z(G).$$

Now consider $a, b \in G$, and note since $Z(G)$ is a subgroup of G we know $G/Z(G)$ forms a partition of G and thus a and b belong to some left coset of $Z(G)$. Therefore by the previous result there exists some $\alpha, \beta \in \mathbb{Z}$ and $z_1, z_2 \in Z(G)$ such that $a = x^\alpha z_1$ and $b = x^\beta z_2$. Since z_1, z_2 are commute with every element in G ,

$$ab = x^\alpha (z_1 x^\beta) z_2 = x^{\alpha+\beta} (z_1 z_2) = x^{\alpha+\beta} z_2 z_1 = x^\beta (x^\alpha z_2) z_1 = x^\beta z_2 x^\alpha z_1 = ba$$

$\square$